

CHAPTER 3

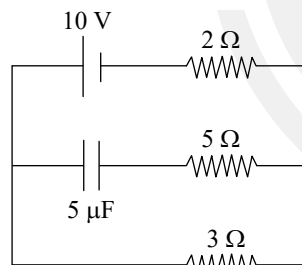
Current Electricity

Current and Motion of Charges

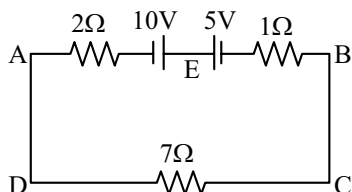
1. A uniform wire of diameter d carries a current of 100 mA when the mean drift velocity of electrons in the wire is v . For a wire of diameter $\frac{d}{2}$ of the same material to carry a current of 200 mA, the mean drift velocity of electrons in the wire is (2024 Re)
- a. $4v$ b. $8v$ c. v d. $2v$

Ohm's Law Resistance and Resistivity

2. In an electrical circuit, the voltage is measured as $V = (200 \pm 4)$ volt and the current is measured as $I = (20 \pm 0.2)A$. The value of the resistance is: (2024 Re)
- a. $(10 \pm 4.2) \Omega$ b. $(10 \pm 0.3) \Omega$
c. $(10 \pm 0.1) \Omega$ d. $(10 \pm 0.8) \Omega$
3. The steady state current in the circuit shown below is: (2024 Re)

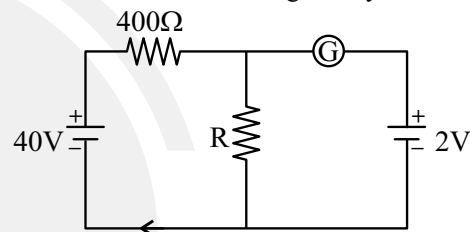


- a. 0.67 A b. 1.5 A c. 2 A d. 1 A
4. A copper wire of radius 1 mm contains 10^{22} free electrons per cubic metre. The drift velocity for free electrons when 10 A current flows through the wire will be (Given, charge on electron = $1.6 \times 10^{-19} C$): (2023-Manipur)
- a. $\frac{6.25 \times 10^4}{\pi} \text{ ms}^{-1}$ b. $\frac{6.25}{\pi} \times 10^3 \text{ ms}^{-1}$
c. $\frac{6.25}{\pi} \text{ ms}^{-1}$ d. $\frac{6.25 \times 10^5}{\pi} \text{ ms}^{-1}$
5. The magnitude and direction of the current in the following circuit is (2023)



- a. 1.5 A from B A through E
b. 0.2 A from B to A through E
c. 0.5 A from A to B through E
d. $\frac{5}{9}$ A from A to B through E

6. If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by: (2023)



- a. 400 Ω b. 200 Ω c. 50 Ω d. 100 Ω
7. The resistance of platinum wire at $0^\circ C$ is 2Ω and 6.8Ω at $80^\circ C$. The temperature coefficient of resistance of the wire is: (2023)
- a. $3 \times 10^{-1} ^\circ C^{-1}$ b. $3 \times 10^{-4} ^\circ C^{-1}$
c. $3 \times 10^{-3} ^\circ C^{-1}$ d. $3 \times 10^{-2} ^\circ C^{-1}$
8. On the basis of electrical conductivity, which one of the following material has the smallest resistivity? (2023-Manipur)
- a. Germanium b. Silver
c. Glass d. Silicon
9. A certain wire A has resistance 81Ω . The resistance of another wire B of same material and equal length but of diameter thrice the diameter of A will be: (2023-Manipur)
- a. 81Ω b. 9Ω c. 729Ω d. 243Ω
10. The reciprocal of resistance is: (2022 Re)
- a. conductance b. reactance
c. mobility d. conductivity
11. Two solid conductors are made up of same material have same length and same resistance. One of them has a circular cross section of area A_1 and the other one has a square cross section of area A_2 . The ratio A_1/A_2 is (2020-Covid)
- a. 1 b. 0.8 c. 2 d. 1.5
12. The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length, its new resistance will be: (2017-Delhi)
- a. $\frac{R}{n}$ b. $n^2 R$ c. $\frac{R}{n^2}$ d. nR

13. Across a metallic conductor of non-uniform cross section a constant potential difference is applied. The quantity which remains constant along the conductor is: (2015)
 a. Current b. Drift velocity
 c. Electric field d. Current density
14. A wire of resistance $4\ \Omega$ is stretched to twice its original length. The resistance of stretched wire would be: (2013)
 a. $16\ \Omega$ b. $2\ \Omega$ c. $4\ \Omega$ d. $8\ \Omega$

Resistance and Conductance

15. A copper wire of length 10 m and radius $(10^{-2}/\sqrt{\pi})\text{ m}$ has electrical resistance of $10\ \Omega$. The current density in the wire for an **electric field** strength of 10 V/m is : (2022)
 a. 10^5 A/m^2 b. 10^4 A/m^2 c. 10^6 A/m^2 d. 10^{-5} A/m^2
16. Column-I gives certain physical terms associated with flow of current through a metallic conductor. Column-II gives some mathematical relations involving electrical quantities. Match Column-I and Column-II with appropriate relations. (2021)

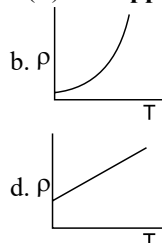
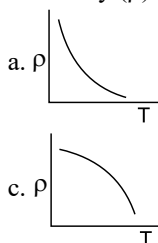
Column-I

- (A) Drift Velocity
 (B) Electrical Resistivity
 (C) Relaxation Period
 (D) Current Density

Column-II

- (P) $\frac{m}{ne^2\rho}$
 (Q) nev_d
 (R) $\frac{eE}{m}\tau$
 (S) $\frac{E}{J}$

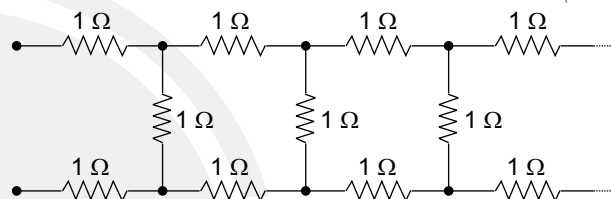
- a. (A) - (R), (B) - (S), (C) - (Q), (D) - (P)
 b. (A) - (R), (B) - (P), (C) - (S), (D) - (Q)
 c. (A) - (R), (B) - (Q), (C) - (S), (D) - (P)
 d. (A) - (R), (B) - (S), (C) - (P), (D) - (Q)
17. A charged particle having drift velocity of $7.5 \times 10^{-4}\text{ m S}^{-1}$ in an electric field of $3 \times 10^{-10}\text{ Vm}^{-1}$, has a mobility in $\text{m}^2\text{ V}^{-1}\text{ S}^{-1}$ of : (2020)
 a. 2.5×10^6 b. 2.5×10^{-6}
 c. 2.25×10^{-15} d. 2.25×10^{15}
18. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for **copper** (2020)



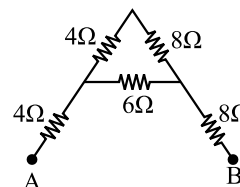
19. Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is: (2015 Re)
- a. $\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$ b. $\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$ c. $\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$ d. $\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$

Combination of Resistance

20. A uniform metal wire of length l has $10\ \Omega$ resistance. Now this wire is stretched to a length $2l$ and then bent to form a perfect circle. The equivalent resistance across any arbitrary diameter of that circle is (2024 Re)
 a. $10\ \Omega$ b. $5\ \Omega$ c. $40\ \Omega$ d. $20\ \Omega$
21. A wire of length ' l ' and resistance $100\ \Omega$ is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is: (2024)
 a. $55\ \Omega$ b. $60\ \Omega$ c. $26\ \Omega$ d. $52\ \Omega$
22. 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is: (2023)
 a. 1000 b. 10 c. 100 d. 1
23. The equivalent resistance of the infinite network given below is: (2022 Re)



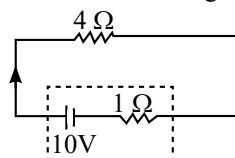
- a. $(1+\sqrt{5})\ \Omega$ b. $2\ \Omega$ c. $(1+\sqrt{2})\ \Omega$ d. $(1+\sqrt{3})\ \Omega$
24. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is $0.25\ \Omega$. What will be the effective resistance if they are connected in series? (2021)
 a. $0.5\ \Omega$ b. $1\ \Omega$ c. $4\ \Omega$ d. $0.25\ \Omega$
25. The equivalent resistance between A and B for the mesh shown in the figure is (2020-Covid)



- a. $16\ \Omega$ b. $30\ \Omega$ c. $4.8\ \Omega$ d. $7.2\ \Omega$
26. A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm , the reading in the ammeter will be: (2015 Re)
 a. 1 A b. 0.5 A c. 0.25 A d. 2 A

Internal Resistance of a Cell and Grouping

27. The terminal voltage of the battery, whose emf is 10 V and internal resistance $1\ \Omega$, when connected through an external resistance of $4\ \Omega$ as shown in the figure is : (2024)



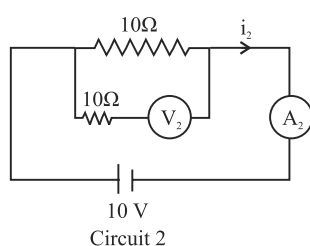
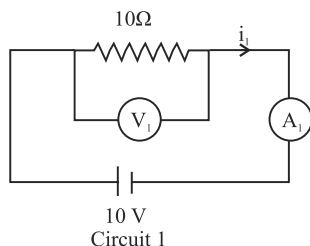
- a. 8 V b. 10 V c. 4 V d. 6 V

28. A cell of emf 4 V and internal resistance 0.5Ω is connected to a 7.5Ω external resistance.

The terminal potential difference of the cell is. (2022 Re)

- a. 0.375 V b. 3.75 V c. 4.25 V d. 4 V

29. In the circuits shown below, the readings of the voltmeters and the ammeters will be (2019)

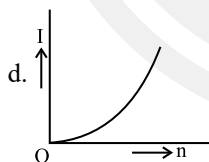
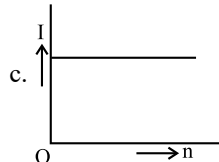
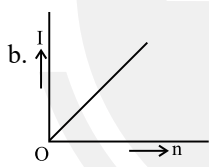
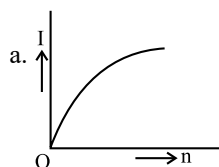


- a. $V_2 > V_1$ and $i_1 = i_2$ b. $V_1 = V_2$ and $i_1 > i_2$
c. $V_1 = V_2$ and $i_1 = i_2$ d. $V_2 > V_1$ and $i_1 > i_2$

30. A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10 I. The value of 'n' is (2018)

- a. 20 b. 11 c. 10 d. 9

31. A battery consists of a variable number 'n' of identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n? (2018)

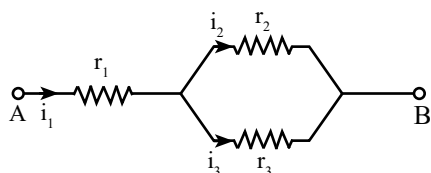


32. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is: (2013)

- a. 1.0Ω b. 0.2Ω c. 0.5Ω d. 0.8Ω

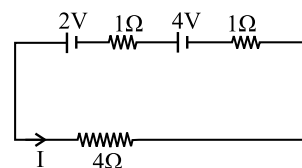
Kirchhoff's Law

33. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is: (2021)



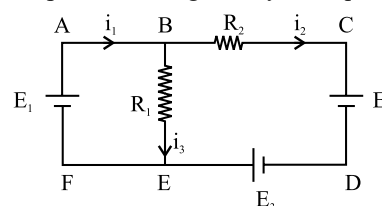
- a. $\frac{r_2}{r_2 + r_3}$ b. $\frac{r_1}{r_1 + r_2}$ c. $\frac{r_2}{r_1 + r_3}$ d. $\frac{r_1}{r_2 + r_3}$

34. For the circuit shown in the figure, the current I will be (2020-Covid)



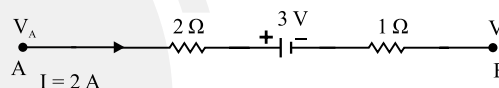
- a. 1 A b. 1.5 A c. 0.5 A d. 0.75 A

35. For the circuit given below, the Kirchoff's loop rule for the loop BCDEB is given by the equation (2020-Covid)



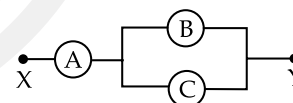
- a. $i_2 R_2 + E_2 - E_3 - i_3 R_1 = 0$
b. $i_2 R_2 + E_2 + E_3 - i_3 R_1 = 0$
c. $-i_2 R_2 + E_2 + E_3 - i_3 R_1 = 0$
d. $-i_2 R_2 + E_2 - E_3 - i_3 R_1 = 0$

36. The potential difference ($V_A - V_B$) between the points A and B in the given figure is: (2016 - II)



- a. +6 V b. +9 V c. -3 V d. +3 V

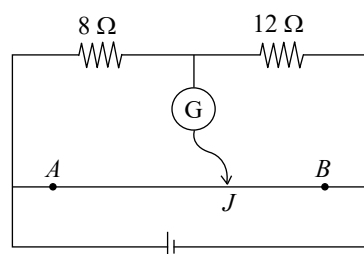
37. A, B and C are voltmeters of resistance R, 1.5R and 3R respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are V_A , V_B and V_C respectively, then: (2015)



- a. $V_A \neq V_B \neq V_C$ b. $V_A = V_B \neq V_C$
c. $V_A \neq V_B \neq V_C$ d. $V_A = V_B = V_C$

Wheatstone and Meter Bridge

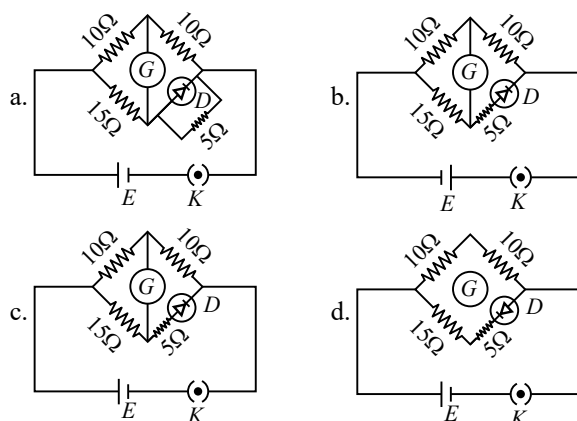
38. The given circuit shows a uniform straight wire AB of 40 cm length fixed at both ends. In order to get zero reading in the galvanometer G, the free end of J is to be placed from B at:



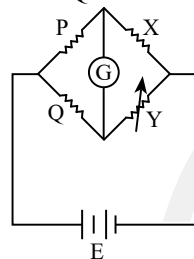
- a. 32 cm b. 8 cm c. 16 cm d. 24 cm

(2024 Re)

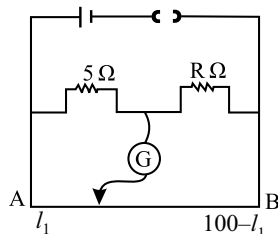
39. Choose the correct circuit which can achieve the bridge balance. (2024)



40. A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X , the resistances P and Q : (2022)



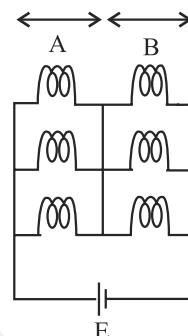
- a. Do not play any significant role
b. Should be approximately equal to $2X$
c. Should be approximately equal and are small
d. Should be very large and unequal
41. A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio $3 : 2$. If the length of the resistance wire is 1.5 m, then the length of 1Ω of the resistance wire is: (2020)
- a. 1.0×10^{-1} m b. 1.5×10^{-1} m
c. 1.5×10^{-2} m d. 1.0×10^{-2} m
42. The resistances in the two arms of the meter bridge are 5Ω and $R\Omega$, respectively. When the resistance R is shunted with an equal resistance, the new balance point is at $1.6 l_1$. The resistance R is: (2014)



- a. 10Ω b. 15Ω c. 20Ω d. 25Ω
43. The resistances of the four arms P , Q , R and S in a Wheatstone's bridge are 10 ohm , 30 ohm , 30 ohm and 90 ohm , respectively. The e.m.f. and internal resistance of the cell are 7 volt and 5 ohm respectively. If the galvanometer resistance is 50 ohm , the current drawn from the cell will be: (2013)
- a. 2.0 A b. 1.0 A c. 0.2 A d. 0.1 A

Electrical Power

44. Two heaters A and B have power rating of 1 kW and 2 kW , respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is: (2024)
- a. $1 : 2$ b. $2 : 3$ c. $1 : 1$ d. $2 : 9$
45. Six similar bulbs are connected as shown in the figure with a DC source of emf E and zero internal resistance. The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be: (2019)



- a. $4 : 9$ b. $9 : 4$ c. $1 : 2$ d. $2 : 1$
46. A filament bulb (500 W , 100 V) is to be used in a 230 V main supply. When a resistance R is connected in series, it works perfectly and the bulb consumes 500 W . The value of R is: (2016 - II)
- a. 26Ω b. 13Ω c. 230Ω d. 46Ω
47. Two cities are 150 km apart. Electric power is sent from one city to another city through copper wires. The fall of potential per km is 8 volt and the average resistance per km is 0.5Ω . The power loss in the wire is: (2014)
- a. 19.2 W b. 19.2 kW c. 19.2 J d. 12.2 kW

Electric Energy and Heating of Current

48. As the temperature **increases**, the electrical resistance: (2022)
- a. decrease for conductors but increases for semiconductors
b. increases for both conductors and semiconductors
c. decreases for both conductors and semiconductors
d. increases for conductors but decreases for semiconductors
49. Two resistors of resistance, 100Ω and 200Ω are connected **in parallel** in an electrical circuit. The ratio of the thermal energy developed in 100Ω to that in 200Ω in a given time is: (2022)
- a. $4 : 1$ b. $1 : 2$ c. $2 : 1$ d. $1 : 4$
50. Which of the following acts as a circuit protection device? (2019)
- a. Conductor b. Inductor
c. Switch d. Fuse
51. The charge flowing through a resistance R varies with time t as $Q = at - bt^2$, where a and b are positive constants. The total heat produced in R is: (2016 - I)

- a. $\frac{a^3 R}{6b}$ b. $\frac{a^3 R}{3b}$ c. $\frac{a^3 R}{2b}$ d. $\frac{a^3 R}{b}$

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (c) | 4. (b) | 5. (c) | 6. (d) | 7. (d) | 8. (b) | 9. (b) | 10. (a) |
| 11. (a) | 12. (b) | 13. (a) | 14. (a) | 15. (a) | 16. (d) | 17. (a) | 18. (b) | 19. (b) | 20. (a) |
| 21. (d) | 22. (c) | 23. (d) | 24. (c) | 25. (a) | 26. (b) | 27. (a) | 28. (b) | 29. (c) | 30. (c) |
| 31. (c) | 32. (c) | 33. (a) | 34. (a) | 35. (a) | 36. (b) | 37. (d) | 38. (d) | 39. (c) | 40. (c) |
| 41. (a) | 42. (b) | 43. (c) | 44. (d) | 45. (b) | 46. (a) | 47. (b) | 48. (d) | 49. (c) | 50. (d) |
| 51. (a) | | | | | | | | | |

